Five reasons to do $\mathcal{L}$ mech $\mathcal{L}(q_i, \dot{q}_i, x)$ or $\mathcal{L}(q_i \dot{q}_i, t)$

① don't need to be in inertial frame

② can handle mass/massless, fields, ---; and skip variational calc

E-L: $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0$ ← came from the principle of stable action

$\delta I = \delta \int_{t_1}^{t_2} \mathcal{L}\, dt$ or

$\delta I = \delta \int_{x_1}^{x_2} \mathcal{L}\, dx$ or.

$\frac{\partial}{\partial t}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} + \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial q_i'} - \frac{\partial \mathcal{L}}{q_i} = 0$ ← ---

$\delta I = \delta \int_{t_1}^{t_2}\int_{x_1}^{x_2} \mathcal{L}\, dx\, dt$ (field theory)

where, $\dot{q}_i = \frac{\partial q_i}{\partial t}$ $q_i' = \frac{\partial q_i}{\partial x}$

③ when deriving E-L, $\dot{q}_i$, q_i', q_i are considered independent, so easier math.

④ can formulate at one coordinates, then transform to the other:

$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = 0$ $\mathcal{L}(q, \dot{q}, t)$

If we have $Q(q, t)$

the we can write: $\mathcal{L}' = \mathcal{L}(Q, \dot{Q}, t)$

$$\frac{d}{dt}\frac{\partial \mathcal{L}'}{\partial \dot{Q}} - \frac{\partial \mathcal{L}'}{\partial Q} = 0$$

(we did the coordinate transfer example before, for coorioles force)

to prove?

$$Q = Q(q,t) \rightarrow \dot{Q} = \frac{d}{dt}Q(q,t) = \frac{\partial Q}{\partial q}\dot{q} + \frac{\partial Q}{\partial t}$$

multi-variable chain rule

$$\frac{\partial \dot{Q}}{\partial \dot{q}} = \frac{\partial Q}{\partial q}$$

$$\frac{\partial \dot{Q}}{\partial q} = \frac{\partial^2 Q}{\partial q^2}\dot{q} + \frac{\partial^2 Q}{\partial q \partial t}$$

$$\frac{\partial \mathcal{L}'}{\partial q} = \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial \dot{Q}}{\partial q} + \frac{\partial \mathcal{L}'}{\partial Q}\frac{\partial Q}{\partial q}$$

$$\frac{\partial \mathcal{L}'}{\partial \dot{q}} = \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial \dot{Q}}{\partial \dot{q}} + \frac{\partial \mathcal{L}'}{\partial Q}\frac{\partial Q}{\partial \dot{q}} = \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial Q}{\partial q}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}'}{\partial \dot{q}} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{Q}}\right)\frac{\partial Q}{\partial q} + \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{d}{dt}\left(\frac{\partial Q}{\partial q}\right)$$

$d(fg) = g\,df + f\,dg$

$$= \frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{Q}}\right)\frac{\partial Q}{\partial q} + \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial \dot{Q}}{\partial q}$$

multivariable chain rule again!

$$\frac{d}{dt}\left(\frac{\partial Q}{\partial q}\right) = \frac{\partial}{\partial Q}\left(\frac{\partial Q}{\partial q}\right)\dot{Q} + \frac{\partial}{\partial \dot{Q}}\left(\frac{\partial Q}{\partial q}\right)\dot{\dot{Q}}$$

$$= \frac{\partial \dot{Q}}{\partial q}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}'}{\partial \dot{q}} - \frac{\partial \mathcal{L}'}{\partial q} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{Q}}\right)\frac{\partial Q}{\partial q} + \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial \dot{Q}}{\partial q} - \frac{\partial \mathcal{L}'}{\partial Q}\frac{\partial Q}{\partial q} - \frac{\partial \mathcal{L}'}{\partial \dot{Q}}\frac{\partial \dot{Q}}{\partial q}$$

$$= \left(\frac{d}{dt}\frac{\partial \mathcal{L}'}{\partial \dot{Q}} - \frac{\partial \mathcal{L}'}{\partial Q}\right)\frac{\partial Q}{\partial q}$$

(5) conservation Laws!

$$\frac{\partial \mathcal{L}}{\partial q} = 0 \xrightarrow{\frac{d}{dt}} \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = 0 \rightarrow \frac{\partial \mathcal{L}}{\partial \dot{q}} = C$$

the action of translation symmetry → conservation of momentum

$$\frac{d\mathcal{L}}{dt} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\dot{q}\right) = 0$$

↑ new term

$$\frac{d}{dt}\left(\mathcal{L} - \frac{\partial \mathcal{L}}{\partial \dot{q}}\dot{q}\right) = 0 \quad \text{or}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\dot{q} - \mathcal{L}\right) = 0$$

the action of Time translation Symmetry

↪ conservation of energy

More example for lesson (2)

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - v(V(x) - \dot{x}A(x,t))$$

↳ constraint:

$V(x)$ → familiar scalar field

$\dot{x}A(x,t)$ → vector potential field

$X = X(t)$

as $A(x,t)$, we can have

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \frac{\partial A}{\partial x}\frac{\partial x}{\partial t}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} + vA \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = m\ddot{x} + v\left(\frac{\partial A}{\partial t} + \frac{\partial A}{\partial x}\underbrace{\frac{\partial x}{\partial t}}_{\dot{x}}\right)$$

$$\frac{\partial \mathcal{L}}{\partial x} = -v\nabla V\frac{\partial V}{\partial x} + v\frac{\partial A}{\partial x}\dot{x}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0 \longrightarrow m\ddot{x} + v\left(\frac{\partial A}{\partial t} + \frac{\partial A}{\partial x}\dot{x}\right) + v\nabla V\frac{\partial V}{\partial x} - v\frac{\partial A}{\partial x}\dot{x}$$

$$= 0$$

$$m\ddot{x} + v\frac{\partial A}{\partial t} + v\nabla V\frac{\partial V}{\partial x} = 0$$

(eq. 7.108) in 1D; 3D for homework

$$\underline{E} = -\nabla V - \frac{\partial A}{\partial t}$$

$$\underline{B} = \nabla \times A \qquad v = \dot{x}$$

$$m\ddot{x} = q(\underline{E} + \dot{x} \times \underline{B}) \quad \text{in 1D}$$

Lorentz force $\underline{F} = q(\underline{E} + \underline{V} \times \underline{B})$

example:

$$\frac{\dot{q}_1^2+\dot{q}_2^2}{2} - V(q_1-q_2)$$

2 particle ($m=1$)



potential energy V is distance dependent

$V' = \frac{\partial V}{\partial q}$ $\dot{p} = \frac{dp}{dt}$

$$\dot{p}_1 = -V'(q_1-q_2) \quad (1)$$

$$\dot{p}_2 = V'(q_1-q_2) \quad (2)$$

(1)+(2) $\dot{p}_1+\dot{p}_2=0$ this cancels, conservation Law

$$\frac{\dot{q}_1^2+\dot{q}_2^2}{2} = -V(aq_1+bq_2)$$

$$\dot{p}_1 = -aV' \quad (1)$$

$$\dot{p}_2 = -bV' \quad (2)$$

(1)+(2) $\dot{p}_1+\dot{p}_2 = -(a+b)V'$ not conserved

but $b\dot{p}_1 = -abV' \quad (3)$

$a\dot{p}_2 = -abV' \quad (4)$

(3)−(4) $b\dot{p}_1 - a\dot{p}_2 = 0$ ← still a conservation law (the top one is the special case of this one, actually)

or $\frac{\partial}{\partial t}(bp_1 - ap_2) = 0$



$\frac{m}{2}\dot{q}_1^2 + \frac{M}{2}\dot{q}_2^2 - V(q_1-q_2)$

def: $-\sqrt{m}\,q_2 = q_4$

$\sqrt{m}\,q_1 = q_3$

$\dot{q}_1^2 = \frac{\dot{q}_3^2}{m}$ $\dot{q}_2^2 = \frac{\dot{q}_4^2}{M}$

$\frac{\dot{q}_3^2}{2} + \frac{\dot{q}_4^2}{2} - V\left(\frac{q_3}{\sqrt{m}} + \frac{q_4}{\sqrt{M}}\right)$

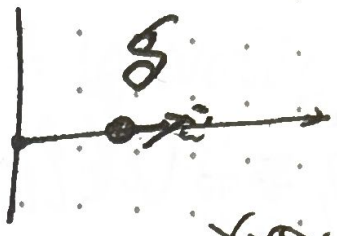
Simplest form of symmetry:

$L = \frac{\dot{q}^2}{2}$ → change of coordinate system or translation (move the subject)

Passive transformation

Active transformation

$\delta q = \delta$



$q' = q + \delta$

$\dot{q}$ doesn't change

$\delta L = 0$

It's a symmetry! / Symmetric.

Translational symmetry, Conservation of momentum conjugative to q

Two particle, continued:

$q_1' = q_1 + b\delta$

$q_2' = q_2 - a\delta$

T doesn't change

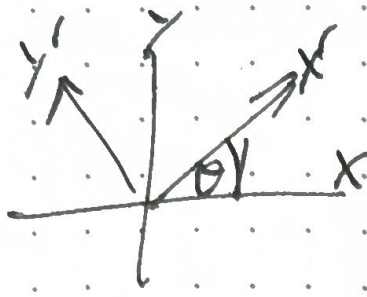
$V(aq_1 + bq_2)$

shift q_1 by $b\delta$: $V(aq_1 + ab\delta + bq_2 - ab\delta)$

$= V(aq_1 + bq_2)$

still doesn't change

ordinary momentum conserved



$\frac{m(\dot{x}^2+\dot{y}^2)}{2} - V(x^2+y^2)$

$x' = \cos\theta\, x + \sin\theta\, y$

$y' = -\sin\theta\, x + \cos\theta\, y$

$\theta = \delta \qquad \cos\theta = 1 \qquad \sin\theta = \delta$

$x' = x + \delta y$ $\qquad$ $\delta y = -x\delta$

$y' = -\delta x + y$ $\qquad$ $\delta x = y\delta$

$\delta\dot{x} = \dot{y}\delta$

$\delta\dot{y} = -\dot{x}\delta$

$\delta(x^2+y^2) = 2x\delta x + 2y\delta y$ ←

$= 2xy\delta - 2xy\delta = 0$

∪ doesn't change;

$\delta(\dot{x}^2+\dot{y}^2) = 0$ T doesn't change!

δ operator → virtual variation.

very much like (d/∂) differential operator;

but $\delta t = 0$

$\delta x^2 = 2x\delta x \qquad \delta xy = x\delta y + y\delta x$

$\delta(f(x)) = \frac{\partial f}{\partial x}\delta x \qquad \delta\dot{x} = \dot{(\delta x)}$